



Student Name: _____

Maths class: _____

James Ruse Agricultural High School

2023 YEAR 12 Task 3 HSC Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 95 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 6–10, show relevant mathematical reasoning and/ or calculations

Total marks: 55

Section I – 5 marks (pages 2–3)

- Attempt Questions 1–5
- Allow about 7 minutes for this section

Section II – 50 marks (pages 4–7)

- Attempt Questions 6–10
- Allow about 1 hour and 28 minutes for this section

Section I

5 marks

Attempt Questions 1-5

Allow about 7 minutes for this section

Write your answers on the Multiple Choice sheet provided

1. Let P be the point $(2, 5, -3)$. What is the distance of P from the yz -plane?
 - A. 2
 - B. $\sqrt{29}$
 - C. $\sqrt{34}$
 - D. $\sqrt{38}$

2. Given that $a > b > 0$, which of the following is not always true?
 - A. $a^2 + b^2 > 2ab$
 - B. $a^2 + b^2 > 4ab$
 - C. $(a + b)^2 > 4ab$
 - D. $a + b > 2\sqrt{ab}$

3. If x, y and z are any real numbers and $x > y$, which of the following statements must be true?
 - A. $x^2 > y^2$
 - B. $\frac{1}{x} < \frac{1}{y}$
 - C. $x + z > y + z$
 - D. $xz > yz$

4. Consider the vector equations of the lines below.

$$\vec{r} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix} \text{ and } \vec{r} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 8 \\ 1 \end{pmatrix}$$

Which of the following statements is correct?

- A. The lines are parallel with no point of intersection.
 - B. The lines are not parallel with no point of intersection.
 - C. The lines are parallel with a point of intersection.
 - D. The lines are not parallel with a point of intersection.
5. Let A, B, C be distinct, non-collinear points in the plane with position vectors $\underline{a}, \underline{b}$ and $\underline{c}$ respectively. P has position vector $\underline{p}$ and is a variable, distinct fourth point in the plane.

Consider the following statements:

- I. $(\underline{p} - \underline{a}) \cdot (\underline{b} - \underline{c}) = 0$
- II. $(\underline{p} - \underline{b}) \cdot (\underline{c} - \underline{a}) = 0$
- III. $(\underline{p} - \underline{c}) \cdot (\underline{a} - \underline{b}) = 0$

Which of the following conclusions can be made about these statements?

- A. Either all statements are true or all statements are false.
- B. If two statements are true, then the third is also true.
- C. Two statements are true if and only if the other one is true.
- D. None of the above.

End of Section I

Section II

50 marks

Attempt Questions 6-10

Allow about 1 hour and 28 minutes for this section

Start each question on a new page. Extra paper is available.

Include relevant mathematical reasoning and/or calculations.

Question 6 (10 marks) Start a new page.

(a) For any integer x , where $x > 1$,

(i) Show that $\frac{1}{x^2} < \frac{1}{x(x-1)}$ 1

(ii) Given that $\frac{1}{x^2-x} = \frac{1}{x-1} - \frac{1}{x}$, show that, for a positive integer n 3

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < 2$$

(b) A sequence of numbers is defined by 3

$$a_0 = 5$$

$$a_n = 2a_{n-1} + 1 \text{ for } n \geq 1$$

Prove by induction that, for integers $n \geq 0$,

$$a_n = 2^{n+2} + 2^{n+1} - 1$$

(c) Find the shortest distance of the point $(1, 2, 3)$ from the line that passes through the origin and the point $(-2, 6, 5)$. 3

Question 7 (10 marks) Start a new page.

(a) Show that $x^4 + y^4 + z^2 + 1 \geq 2x(xy^2 - x + z + 1)$ 2

(b) A particle moves such that at any time t , its position vector is given by $\underline{r} = \begin{pmatrix} -2t^2 + 3t \\ 4t^2 \\ 6t - 4 \end{pmatrix}$. 2

Find the speed of the particle at time after 3 seconds.

(c) The sphere $\mathcal{S}$ has equation $(x - 2)^2 + (y + 1)^2 + (z + 3)^2 = 169$.

(i) Write equation of $\mathcal{S}$ in vector form, stating its centre and radius. 2

(ii) A line segment in three-dimensional space has equation 4

$$\underline{r} = (7\underline{i} + \underline{j} + 12\underline{k}) + \lambda(2\underline{i} - 2\underline{j} + 3\underline{k}) \quad \text{for } 0 < \lambda < 3.$$

Find the point(s) of intersection between this line segment and the sphere $\mathcal{S}$.

Question 8 (10 marks) Start a new page.

(a) A function $f(x)$ is defined such that $f(x) > 0$ for all real numbers x and

$$f(a + b) = f(a) \times f(b) \text{ for any real numbers } a \text{ and } b.$$

(i) Show that $f(0) = 1$ and deduce that $f(-x) = \frac{1}{f(x)}$ 3

(ii) By using mathematical induction, show that $f(nx) = [f(x)]^n$ for $n \in \mathbb{Z}^+$. 2

(b) Let $\underline{a}$ and $\underline{b}$ be the position vectors of the fixed points A and B respectively and let $\underline{v}$ be the position vector of the variable point P . 3

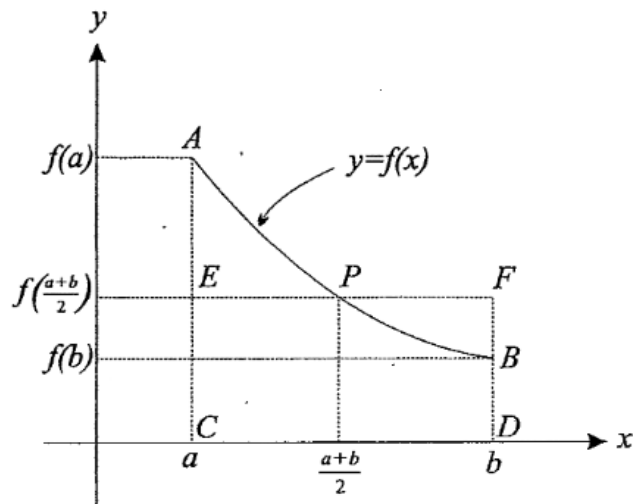
(i) Given that $|\underline{a} - \underline{v}| = |\underline{b} - \underline{v}|$, find the value of $(\underline{b} - \underline{a}) \cdot \left(\underline{v} - \frac{\underline{a} + \underline{b}}{2}\right)$. 1

(ii) If $\underline{a}$, $\underline{b}$ and $\underline{v}$ are vectors in the plane, give a geometrical interpretation for the set of all possible points P . 1

(iii) Hence, geometrically interpret the set of all possible points P if $\underline{a}$, $\underline{b}$ and $\underline{v}$ are vectors in three-dimensional space.

Question 9 (10 marks) Start a new page.

- (a) The diagram below shows the curve $y = f(x)$ for $a \leq x \leq b$. It is given that $f(x)$ is concave up for $a \leq x \leq b$.



- (i) Use areas to explain why

3

$$(b-a)f\left(\frac{a+b}{2}\right) < \int_a^b f(x) dx < (b-a) \frac{f(a) + f(b)}{2}$$

- (ii) Let $f(x) = \frac{1}{x^2}$, $a = n-1$ and $b = n$, where $n \in \mathbb{Z}$, $n > 1$. Hence show that

2

$$\frac{4}{(2n-1)^2} < \frac{1}{n-1} - \frac{1}{n} < \frac{1}{2} \left(\frac{1}{(n-1)^2} + \frac{1}{n^2} \right)$$

- (iii) Deduce that

2

$$4 \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots \right) < 1 < \frac{1}{2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$$

- (iv) Show that

1

$$\frac{1}{2} \left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots \right) < \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

- (v) Hence show that

2

$$\frac{3}{2} < \sum_{n=1}^{\infty} \frac{1}{n^2} < \frac{7}{4}$$

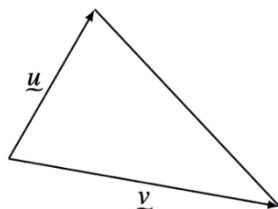
Question 10 (10 marks) **Start a new page.**

- (a) Find the Cartesian equation of the curve defined parametrically by $\underline{r} = \begin{pmatrix} \cos t (1 + \tan t) \\ \sin t (1 + \tan t) \end{pmatrix}$,
stating any restrictions necessary.

4

- (b) (i) The triangle formed by vectors $\underline{u}$ and $\underline{v}$ is shown below.

3

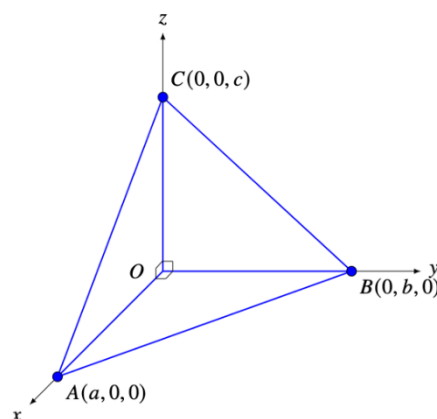


By considering the projection of $\underline{u}$ onto $\underline{v}$, show that the area of the triangle is given by

$$A = \frac{1}{2} \sqrt{|\underline{u}|^2 |\underline{v}|^2 - (\underline{u} \cdot \underline{v})^2}.$$

- (ii) Now consider the tetrahedron $OABC$ below with points A , B and C lying on the x , y and z axes respectively.

3



Let $|XYZ|$ denote the area of ΔXYZ . Using vector methods and the result in part (i), show that

$$|AOB|^2 + |AOC|^2 + |BOC|^2 = |ABC|^2.$$

END OF PAPER

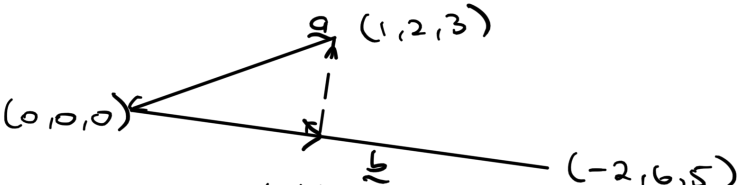
MATHEMATICS Extension 2: Question...6..

Suggested Solutions	Marks	Marker's Comments
a) i) RTP: $\frac{1}{x^2} < \frac{1}{x(x-1)}$ ie. $\frac{1}{x(x-1)} - \frac{1}{x^2} > 0$ LHS = $\frac{x - (x-1)}{x^2(x-1)}$ = $\frac{1}{x^2(x-1)}$ > 0 as $x^2 > 0$ and $x > 1$ $\therefore x-1 > 0$ OR $x > x-1$ as $x > 1$ $x^2 > x^2 - x$ $x^2 > x(x-1)$ $\frac{1}{x^2} < \frac{1}{x(x-1)}$ as $x > 1$ $x^2 > 0$ $x-1 > 0$ ii) $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 2$ LHS = $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2}$ < $\frac{1}{1^2} + \frac{1}{2(2-1)} + \frac{1}{3(3-1)} + \frac{1}{4(4-1)}$ $+ \dots + \frac{1}{n(n-1)}$ from (i) = $1 + \frac{1}{2-1} - \frac{1}{2} + \frac{1}{3-1} - \frac{1}{3}$ $+ \frac{1}{4-1} - \frac{1}{4} + \dots + \frac{1}{n-1} - \frac{1}{n}$ given $\frac{1}{x^2-x} = \frac{1}{x-1} - \frac{1}{x}$ = $1 + 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n-1} - \frac{1}{n}$	1	I must have reasoning to support your claim.

MATHEMATICS Extension 2: Question...6..

Suggested Solutions	Marks	Marker's Comments
$= 2 - \frac{1}{n}$ $< 2 \quad \text{as } n \in \mathbb{Z}^+ \quad \therefore \frac{1}{n} > 0$ b) $a_0 = 5$ $a_n = 2a_{n-1} + 1$ $a_n = 2^{n+2} + 2^{n+1} - 1$ for $n=0$ $a_0 = 2^{0+2} + 2^{0+1} - 1$ $= 4 + 2 - 1$ $= 5$ $\therefore$ true for $n=0$ Assume true for $n=k, k \in \mathbb{Z}, k \geq 0$ ie. $a_k = 2^{k+2} + 2^{k+1} - 1$ Prove true for $n=k+1$ ie. $a_{k+1} = 2^{(k+1)+2} + 2^{(k+1)+1} - 1$ LHS = a_{k+1} $= 2a_k + 1$ (by definition) $= 2(2^{k+2} + 2^{k+1} - 1) + 1$ (by assumption) $= 2 \times 2^{k+2} + 2 \times 2^{k+1} - 2 + 1$ $= 2^{(k+1)+2} + 2^{(k+1)+1} - 1$ $= \text{RHS}$ $\therefore$ true for $n=k+1$ $\therefore$ the statement is true by the principle of mathematical induction.	1 1 1 1	need to have reasoning 1 for setting out of induction question - $k \in \mathbb{Z}, k \geq 0$ - statement for $n=k$ - statement for $n=k+1$ - by assumption - conclusion

MATHEMATICS Extension 2: Question...6..

Suggested Solutions	Marks	Marker's Comments
c)  let $\underline{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ $\underline{b} = \begin{pmatrix} -2 \\ 6 \\ 5 \end{pmatrix}$ $\therefore$ shortest distance $= \text{proj}_{\underline{b}} \underline{a} - \underline{a} $ $= \left \left(\frac{\underline{a} \cdot \underline{b}}{ \underline{b} ^2} \right) \underline{b} - \underline{a} \right $ $= \left \left(\frac{-2 + 12 + 15}{4 + 36 + 25} \right) \begin{pmatrix} -2 \\ 6 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right $ $= \left \frac{5}{13} \begin{pmatrix} -2 \\ 6 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right $ $= \left \begin{pmatrix} -\frac{23}{13} \\ \frac{4}{13} \\ -\frac{14}{13} \end{pmatrix} \right $ $= \sqrt{\left(-\frac{23}{13}\right)^2 + \left(\frac{4}{13}\right)^2 + \left(-\frac{14}{13}\right)^2}$ $= \sqrt{\frac{57}{13}}$	1 1 1	correct proj $\underline{a}$

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

OR

let the line passing through origin and point $(-2, 6, 5)$ be:

$$\underline{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 6 \\ 5 \end{pmatrix} \text{ where } \lambda \in \mathbb{R}$$

$\therefore$ any point can be written as:

$$\begin{pmatrix} -2\lambda \\ 6\lambda \\ 5\lambda \end{pmatrix}$$

$\therefore$ perpendicular vector through $\underline{r}$ and $(1, 2, 3)$ is:

$$\begin{pmatrix} -2\lambda - 1 \\ 6\lambda - 2 \\ 5\lambda - 3 \end{pmatrix}$$

as they are perpendicular:

$$\begin{pmatrix} -2\lambda - 1 \\ 6\lambda - 2 \\ 5\lambda - 3 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 6 \\ 5 \end{pmatrix} = 0$$

$$(-2\lambda - 1)(-2) + (6\lambda - 2)(6) + (5\lambda - 3)(5) = 0$$

$$4\lambda + 2 + 36\lambda - 12 + 25\lambda - 15 = 0$$

$$65\lambda = 25$$

$$\lambda = \frac{5}{13}$$

$$\therefore \text{ vector is: } \begin{pmatrix} -2\left(\frac{5}{13}\right) - 1 \\ 6\left(\frac{5}{13}\right) - 2 \\ 5\left(\frac{5}{13}\right) - 3 \end{pmatrix} = \begin{pmatrix} -\frac{23}{13} \\ \frac{4}{13} \\ -\frac{14}{13} \end{pmatrix}$$

$\therefore$ shortest distance is:

$$\sqrt{\left(-\frac{23}{13}\right)^2 + \left(\frac{4}{13}\right)^2 + \left(-\frac{14}{13}\right)^2} = \sqrt{\frac{57}{13}}$$

Year 12 Ext. 2 Task 3 Q7 solutions

(a) Method I

$$x^4 + y^4 \geq 2x^2y^2 \quad (1) \text{ (AM-GM)}$$

$$x^2 + z^2 \geq 2xz \quad (2)$$

$$x^2 + 1 \geq 2x \quad (3)$$

1 mark

$$(1) + (2) + (3) : x^4 + y^4 + 2x^2 + z^2 + 1 \geq 2x^2y^2 + 2xz + 2x$$

$$x^4 + y^4 + z^2 + 1 \geq 2x^2y^2 - 2x^2 + 2xz + 2x$$

$$\therefore x^4 + y^4 + z^2 + 1 \geq 2x(xy^2 - x + z + 1)$$

1 mark

Method II

$$(x^2 - y^2)^2 + (x - z)^2 + (x - 1)^2 \geq 0$$

1 mark

$$x^4 - 2x^2y^2 + y^4 + x^2 - 2xz + z^2 + x^2 - 2x + 1 \geq 0$$

$$x^4 + y^4 + z^2 + 1 \geq 2x^2y^2 - 2x^2 + 2xz + 2x$$

$$\therefore x^4 + y^4 + z^2 + 1 \geq 2x(xy^2 - x + z + 1)$$

1 mark

(b)

$$\underline{r} = \begin{pmatrix} -2t^2 + 3t \\ 4t^2 \\ 6t - 4 \end{pmatrix}$$

$$\underline{v} = \begin{pmatrix} -4t + 3 \\ 8t \\ 6 \end{pmatrix}$$

1 mark for finding velocity vector correctly

$$\underline{v}(3) = \begin{pmatrix} -9 \\ 24 \\ 6 \end{pmatrix}$$

$$\therefore \text{speed} = \sqrt{(-9)^2 + (24)^2 + (6)^2}$$

$$= \sqrt{693}$$

$$= 3\sqrt{77}$$

1 mark for correct working and answer

(c)

$$(x-2)^2 + (y+1)^2 + (z+3)^2 = 169$$

i, centre = $(2, -1, -3)$

1 mark for finding BOTH centre and radius correctly

radius = 13

∴ vector form : $\left| \mathbf{x} - \begin{pmatrix} 2 \\ -1 \\ -3 \end{pmatrix} \right| = 13$ 1 mark for correct vector form of sphere

ii) $\mathbf{r} = \begin{pmatrix} 7 \\ 1 \\ 12 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix} \quad 0 < \lambda < 3$

Sub into sphere S

$$\left| \begin{pmatrix} 7 \\ 1 \\ 12 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ -3 \end{pmatrix} \right| = 13$$
 1 mark for substituting correctly

$$\left| \begin{pmatrix} 5 \\ 2 \\ 15 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix} \right| = 13$$

$$(5+2\lambda)^2 + (2-2\lambda)^2 + (15+3\lambda)^2 = 169$$
 1 mark for getting a quadratic equation of lambda

$$(25 + 20\lambda + 4\lambda^2) + (4 - 8\lambda + 4\lambda^2) + (225 + 90\lambda + 9\lambda^2) = 169$$

$$17\lambda^2 + 102\lambda + 85 = 0$$

$$\lambda^2 + 6\lambda + 5 = 0$$

$$(\lambda + 1)(\lambda + 5) = 0$$

$$\therefore \lambda = -1, -5$$

1 mark for correctly solving the quadratic equation

but $0 < \lambda < 3$

∴ no point of intersection

1 mark for rejection of solutions and correct conclusion

Question 8

(a) $f(x) > 0$ for all $x \in \mathbb{R}$

$f(a+b) = f(a) \times f(b)$ for all $a \in \mathbb{R}$ and $b \in \mathbb{R}$

(i) show that $f(0) = 1$ and deduce that $f(-x) = \frac{1}{f(x)}$

$$f(0+0) = f(0) \times f(0)$$

$$f(0) = f(0)^2$$

$$f(0)(1 - f(0)) = 0 \quad \text{but } f(x) > 0 \text{ for all } x$$

$$\therefore f(0) \neq 0 \text{ and } 1 - f(0) = 0 \therefore f(0) = 1$$

$$f(x + (-x)) = f(x) \times f(-x)$$

$$f(0) = f(x) \times f(-x)$$

$$1 = f(x) \times f(-x)$$

$$f(-x) = \frac{1}{f(x)}$$

(ii) By Mathematical induction, show that $f(nx) = [f(x)]^n$ for $n \in \mathbb{Z}^+$

$$\text{for } n=1 \quad f(x) = f(x)^1 = f(x)$$

True for $n=1$

~~for~~ Assume it is true for $k \in \mathbb{Z}^+$

$$f(kx) = [f(x)]^k \quad (*)$$

and prove it true for $n = k+1$

$$\text{i.e. } f((k+1)x) = [f(x)]^{k+1}$$

$$\text{Now } f((k+1)x) = f(kx+x) = f(kx) \times f(x)$$

$$\text{But } f(kx) = [f(x)]^k$$

$$\text{Hence } f((k+1)x) = [f(x)]^k \times f(x) = [f(x)]^{k+1}$$

By mathematical induction $f(nx) = [f(x)]^n$ for all $n \in \mathbb{Z}$

(b) $\underline{a}$ and $\underline{b}$ position vectors of A and B
 $\underline{v}$ position vector of P

(i) Given that $|\underline{a} - \underline{v}| = |\underline{b} - \underline{v}|$

Find the value of

$$(\underline{b} - \underline{a}) \cdot \left(\underline{v} - \frac{\underline{a} + \underline{b}}{2}\right)$$

$$(\underline{b} - \underline{a}) \cdot \left(\underline{v} - \frac{\underline{a} + \underline{b}}{2}\right) = (\underline{b} - \underline{a}) \cdot \underline{v} - (\underline{b} - \underline{a}) \cdot \frac{(\underline{a} + \underline{b})}{2} \quad \checkmark$$

$$|\underline{a} - \underline{v}| = |\underline{b} - \underline{v}| \quad \therefore |\underline{a} - \underline{v}|^2 = |\underline{b} - \underline{v}|^2$$

$$\therefore (\underline{a} - \underline{v}) \cdot (\underline{a} - \underline{v}) = (\underline{b} - \underline{v}) \cdot (\underline{b} - \underline{v}) \quad \checkmark$$

$$\underline{a} \cdot \underline{a} - 2\underline{a} \cdot \underline{v} + \underline{v} \cdot \underline{v} = \underline{b} \cdot \underline{b} - 2\underline{b} \cdot \underline{v} + \underline{v} \cdot \underline{v}$$

$$2(\underline{b} - \underline{a}) \cdot \underline{v} - (\underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{a}) = 0$$

$$\therefore (\underline{b} - \underline{a}) \cdot \underline{v} - \frac{1}{2}(\underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{a}) = 0 \quad \checkmark$$

Note.

$$\begin{aligned} (\underline{b} - \underline{a}) \cdot (\underline{a} + \underline{b}) &= \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{a} - \underline{a} \cdot \underline{b} \\ &= \underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{a} \end{aligned}$$

(ii) $\underline{a}$, $\underline{b}$ and $\underline{v}$ are vectors in the plane, give a geometrical interpretation for the set of all possible points P.

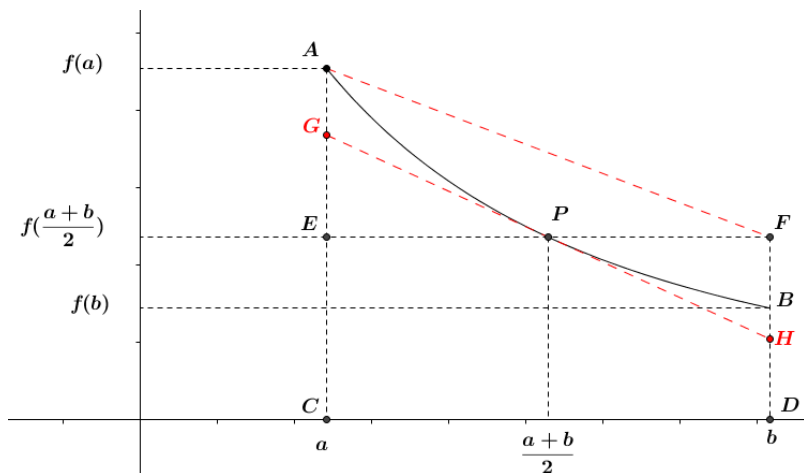
P lies on the perpendicular bisector of AB. $\checkmark$

(iii) Hence, geometrically interpret the set of all possible points P if $\underline{a}$, $\underline{b}$ and $\underline{v}$ are vectors in 3D space.

P lies on the plane which passes by the midpoint of AB and $\vec{AB}$ is Normal to the plane. $\checkmark$

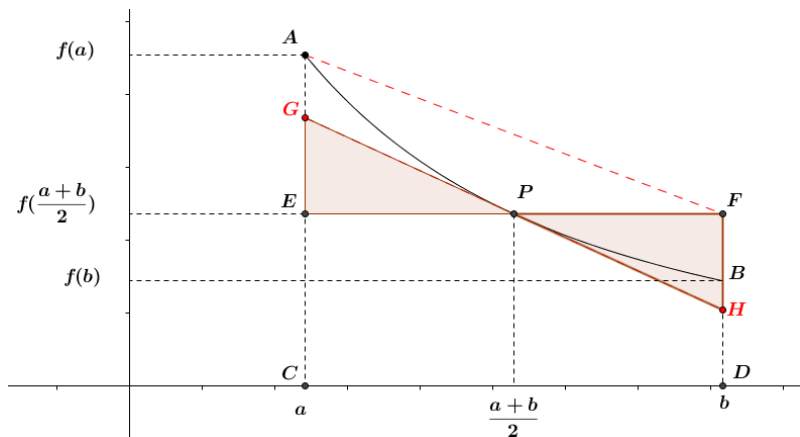
Question 9

- i. Construct a tangent at P to meet AC and FD at G and H respectively.



$$\text{Area Trapezium}_{GHDC} < \text{Area under } y = f(x) \text{ between } x = a \text{ and } x = b < \text{Area Trapezium}_{AFDC}$$

$$\Delta GEP \equiv \Delta HFP \Rightarrow \text{Area}_{\Delta GEP} = \text{Area}_{\Delta HFP}$$

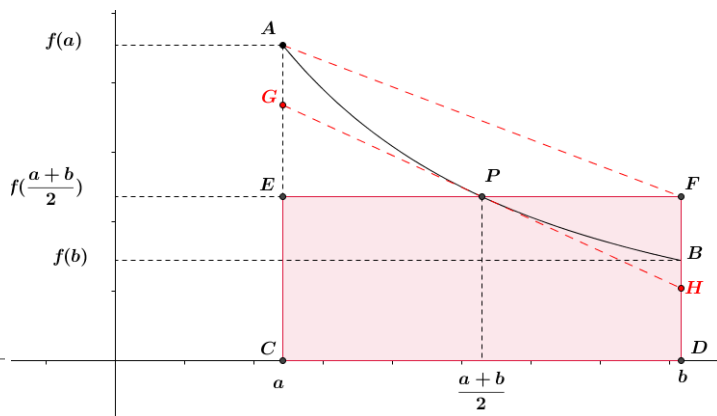
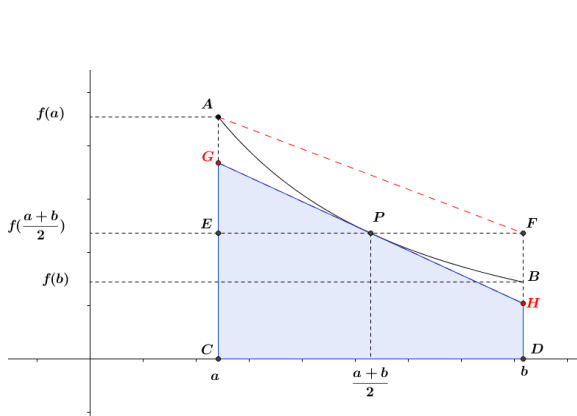


1 mark for identifying what each expression represents in terms of area (i.e. Rectangle, Trapezium and area under the curve).

1 mark for establishing one side of the inequality with sufficient reasons.

1 mark for establishing the other side of the inequality with sufficient reasons.

$$\therefore \text{Area rectangle}_{ECDF} = \text{Area Trapezium}_{GHDC}$$



$$\text{Area rectangle}_{ECDF} < \text{Area under } y = f(x) \text{ between } x = a \text{ and } x = b < \text{Area Trapezium}_{AFDC}$$

$$EC \times CD < \int_a^b f(x) dx < \frac{1}{2}(AC + FD) \times CD$$

$$(b - a)f\left(\frac{a+b}{2}\right) < \int_a^b f(x) dx < (b - a) \times \frac{f(a) + f(b)}{2}$$

Note:

- If you define $(b - a) \times \frac{f(a) + f(b)}{2}$ as the area of a rectangle (or 2 rectangles) you're almost certain to get 0.
- "The curve is concave up" argument is only sufficient to prove $\int_a^b f(x) dx < \text{Area Trapezium}_{GHDC}$

ii.

$$f(x) = \frac{1}{x^2}, a = n - 1, b = n, n > 1, n \in \mathbb{Z}$$

$$\therefore [n - (n - 1)] \times \frac{1}{\left(n - \frac{1}{2}\right)^2} < \int_{n-1}^n \frac{1}{x^2} dx < [n - (n - 1)] \times \frac{\frac{1}{n^2} + \frac{1}{(n-1)^2}}{2}$$

$$\frac{1}{\left(n - \frac{1}{2}\right)^2} < \left[-\frac{1}{x}\right]_{n-1}^n < \frac{1}{2} \times \left(\frac{1}{(n-1)^2} + \frac{1}{n^2}\right)$$

1 mark for getting the middle expression through integration.

$$4 \times \frac{1}{(2n-1)^2} < \left[\frac{1}{x}\right]_n^{n-1} < \frac{1}{2} \left[\frac{1}{(n-1)^2} + \frac{1}{n^2}\right]$$

1 mark for establishing the other two expressions by substitution and showing all necessary working out.

$$\frac{4}{(2n-1)^2} < \frac{1}{n-1} - \frac{1}{n} < \frac{1}{2} \left[\frac{1}{(n-1)^2} + \frac{1}{n^2}\right]$$

iii.

$$n = 2 \Rightarrow \frac{4}{3^2} < 1 - \frac{1}{2} < \frac{1}{2} \left[\frac{1}{1^2} + \frac{1}{2^2}\right] - \text{Equation 1}$$

$$n = 3 \Rightarrow \frac{4}{5^2} < \frac{1}{2} - \frac{1}{3} < \frac{1}{2} \left[\frac{1}{2^2} + \frac{1}{3^2}\right] - \text{Equation 2}$$

$$n = 4 \Rightarrow \frac{4}{7^2} < \frac{1}{4} - \frac{1}{3} < \frac{1}{2} \left[\frac{1}{3^2} + \frac{1}{4^2}\right] - \text{Equation 3}$$

:

$$\therefore \sum_{n=2}^{\infty} \frac{4}{(2n-1)^2} < \sum_{n=2}^{\infty} \frac{1}{n-1} - \frac{1}{n} < \sum_{n=2}^{\infty} \frac{1}{2} \left[\frac{1}{(n-1)^2} + \frac{1}{n^2}\right]$$

$$\frac{4}{3^2} + \frac{4}{5^2} + \frac{4}{7^2} + \dots < \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots < \frac{1}{2} \left[\frac{1}{1^2} + \frac{1}{2^2}\right] + \frac{1}{2} \left[\frac{1}{2^2} + \frac{1}{3^2}\right] + \frac{1}{2} \left[\frac{1}{3^2} + \frac{1}{4^2}\right] + \dots$$

$$4 \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) < \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) < \frac{1}{2} \left[\frac{1}{1^2} + \frac{1}{2^2}\right] + \frac{1}{2} \left[\frac{1}{2^2} + \frac{1}{3^2}\right] + \frac{1}{2} \left[\frac{1}{3^2} + \frac{1}{4^2}\right] + \dots$$

$$4 \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) < 1 < \frac{1}{2} \left(1 + 2 \left(\frac{1}{2^2}\right) + 2 \left(\frac{1}{3}\right)^2 + \dots\right)$$

$$4 \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) < 1 < \frac{1}{2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

1 mark for establishing a pattern correctly starting from $n = 2$ for all 3 expressions.

1 mark for correct addition of equations AND clear working out. You **MUST SHOW** necessary working in addition to just stating "sum all the equations together" before getting to the required expression.

iv.

$$\begin{aligned} \frac{1}{2} \left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots\right) &= \frac{1}{2} \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) + \frac{1}{2} \left(\frac{1}{4^2} + \frac{1}{6^2} + \frac{1}{8^2} + \dots\right) \\ &< \frac{1}{2} \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) + \frac{1}{2} \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) \left(\text{since } \frac{1}{(n-1)^2} > \frac{1}{n^2} \forall n > 1\right) \\ &= \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots \end{aligned}$$

$$\therefore \frac{1}{2} \left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots\right) < \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right)$$

1 mark for getting to required line showing all necessary working.

v.

$$4\left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) < 1 < \frac{1}{2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \quad (\text{proven in iii})$$

$$1 < \frac{1}{2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \Rightarrow \left(1 + \frac{1}{2}\right) < \frac{1}{2} + \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots\right)$$

$$\frac{3}{2} < 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

$$= \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

1 mark for getting the inequality on the left with necessary working out.

$$\therefore \frac{3}{2} < \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$4\left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) < 1 \Rightarrow \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) < \frac{1}{4} \quad (\text{Proven in iii.})$$

$$\frac{1}{2}\left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots\right) < \frac{1}{4} \quad \left(\frac{1}{2}\left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots\right) < \left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots\right) \text{ proven in iv.}\right)$$

$$\left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots\right) < \frac{1}{2}$$

$$\left(\frac{1}{1^2} + \frac{1}{2^2}\right) + \left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots\right) < \frac{1}{2} + \left(\frac{1}{1^2} + \frac{1}{2^2}\right)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < 1 + \frac{1}{2} + \frac{1}{4}$$

$$= \frac{7}{4}$$

1 mark for getting the inequality on the right with necessary working out.

$$\therefore \frac{3}{2} < \sum_{n=1}^{\infty} \frac{1}{n^2} < \frac{7}{4}$$

MATHEMATICS EXT 2: Question 10

Suggested Solutions	Marks	Marker's Comments
a, $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos t (1 + \tan t) \\ \sin t (1 + \tan t) \end{pmatrix}$ $x = \cos t (1 + \tan t)$ ① $y = \sin t (1 + \tan t)$ ② ② ÷ ①: $\frac{y}{x} = \tan t$ ① ①² + ②²: $x^2 + y^2 = \cos^2 t (1 + \tan t)^2 + \sin^2 t (1 + \tan t)^2$ $= (1 + \tan t)^2 (\cos^2 t + \sin^2 t)$ $x^2 + y^2 = (1 + \tan t)^2$ ① $\therefore x^2 + y^2 = \left(1 + \frac{y}{x}\right)^2$ ① But as stated above, (0,0) is not included. However, subbing $t = \frac{3\pi}{4}$ into the original parametric eqn shows that (0,0) is valid, so we must express our cartesian eqn in such a way that it includes (0,0). $\therefore x^2(x^2 + y^2) = (x + y)^2$ ① Notes: - Learn the common tricks for parametrics → squaring & adding → multiplying / dividing → make use of trig identities. - Few students got ③ marks, let alone ④.		Eliminating t completely Including origin.

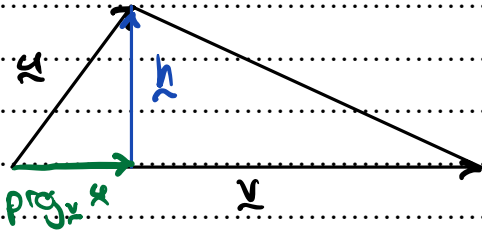
MATHEMATICS EXT 2: Question 10

Suggested Solutions

Marks

Marker's Comments

b),



$$A = \frac{1}{2} b h_\perp$$

$$= \frac{1}{2} |\underline{v}| |h|$$

$$A = \frac{1}{2} |\underline{v}| |\text{proj}_{\underline{v}^\perp} \underline{u}| \quad \text{①}$$

$$A^2 = \frac{1}{4} |\underline{v}|^2 \left| \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \underline{v} - \underline{u} \right|^2$$

$$= \frac{1}{4} |\underline{v}|^2 \left(\frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \underline{v} - \underline{u} \right) \cdot \left(\frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \underline{v} - \underline{u} \right) \quad \text{①}$$

$$= \frac{1}{4} |\underline{v}|^2 \left(\left(\frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \right)^2 \underline{v} \cdot \underline{v} - \frac{(\underline{u} \cdot \underline{v})^2}{|\underline{v}|^2} - \frac{(\underline{u} \cdot \underline{v})^2}{|\underline{v}|^2} + \underline{u} \cdot \underline{u} \right)$$

$$= \frac{1}{4} |\underline{v}|^2 \left(\frac{(\underline{u} \cdot \underline{v})^2}{|\underline{v}|^4} |\underline{v}|^2 - 2 \frac{(\underline{u} \cdot \underline{v})^2}{|\underline{v}|^2} + |\underline{u}|^2 \right)$$

$$= \frac{1}{4} |\underline{v}|^2 \left(|\underline{u}|^2 - \frac{(\underline{u} \cdot \underline{v})^2}{|\underline{v}|^2} \right)$$

$$= \frac{1}{4} \left(|\underline{u}|^2 |\underline{v}|^2 - (\underline{u} \cdot \underline{v})^2 \right)$$

$$\therefore A = \frac{1}{2} \sqrt{|\underline{u}|^2 |\underline{v}|^2 - (\underline{u} \cdot \underline{v})^2} \quad \text{①}$$

Getting to final with sufficient steps shown.

MATHEMATICS EXT 2: Question 10

Suggested Solutions

Marks

Marker's Comments

ii, $\triangle AOB$ is formed by $\vec{OA}$ and $\vec{OB}$

$$|AOB| = \frac{1}{2} \sqrt{|\vec{OA}|^2 |\vec{OB}|^2 - (\vec{OA} \cdot \vec{OB})^2}$$

$$= \frac{1}{2} \sqrt{a^2 b^2 - 0}$$

$$|AOB|^2 = \frac{1}{4} a^2 b^2$$

similarly, $|AOC|^2 = \frac{1}{4} a^2 c^2$ & $|BOC|^2 = \frac{1}{4} b^2 c^2$

$\triangle ABC$ is formed by $\vec{AC}$ and $\vec{AB}$

$$|ABC| = \frac{1}{2} \sqrt{|\vec{AC}|^2 |\vec{AB}|^2 - (\vec{AC} \cdot \vec{AB})^2}$$

$$= \frac{1}{2} \sqrt{(a^2 + c^2)(a^2 + b^2) - a^4}$$

$$|ABC|^2 = \frac{1}{4} (a^4 + a^2 b^2 + a^2 c^2 + b^2 c^2 - a^4)$$

$$= \frac{1}{4} (a^2 b^2 + a^2 c^2 + b^2 c^2)$$

$$= \frac{1}{4} a^2 b^2 + \frac{1}{4} a^2 c^2 + \frac{1}{4} b^2 c^2$$

$$|ABC|^2 = |AOB|^2 + |AOC|^2 + |BOC|^2$$

Notes:

- If you didn't MEANINGFULLY use the projection, no marks awarded
- Too many students confused themselves by introducing a, b, c without defn. They confused the vectors with already defined scalars a, b, c .
- You need to SHOW non-trivial steps!

①

① progress

①

Getting to final.